4.6 The Bernoulli and Binomial Random Variables

Suppose that a trial, or an experiment, whose outcome can be classified as either a *success* or a *failure* is performed. If we let X=1 when the outcome is a success and X=0 when it is a failure, then the probability mass function of X is given by

$$p(0)=P\{ X=0 \}=1-pp(1)=P\{ X=1 \}=p \tag{6.1}$$

where p, $0 \le p \le 1$, is the probability that the trial is a success.

A random variable X is said to be a *Bernoulli random variable* (after the Swiss mathematician James Bernoulli) if its probability mass function is given by **Equations (6.1)** for some $p \in (0,1)$.

Suppose now that n independent trials, each of which results in a success with probability p or in a failure with probability 1−p, are to be performed. If X represents the number of successes that occur in the n trials, then X is said to be a *binomial random variable* with parameters (n, p). Thus, a Bernoulli random variable is just a binomial random variable with parameters (1, p).

The probability mass function of a binomial random variable having parameters (n, p) is given by

$$p(i)=(ni)pi(1-p)n-ii=0,1,\ldots,n \tag{6.2}$$

The validity of **Equation (6.2)** may be verified by first noting that the probability of any particular sequence of n outcomes containing i successes and n−i failures is, by the assumed independence of trials, pi(1−p)n−i. **Equation (6.2)** then follows, since there are (ni) different sequences of the n outcomes leading to i successes and n−i failures. This perhaps can most easily be seen by noting that there are (ni) different choices of the i trials that result in successes. For instance, if n=4,i=2, then there are (42)=6 ways in which the four trials can result in two successes, namely, any of the outcomes (s, s, f, f), (s, f, s, f), (s, f, f, s), (f, s, s, f), (f, s, f, s), and (f, f, s, s), where the outcome (s, s, f, f) means, for instance, that the first two trials are successes and the last two failures. Since each of these outcomes has probability p2(1−p)2 of occurring, the desired probability of two successes in the four trials is (42)p2(1−p)2.

Note that, by the binomial theorem, the probabilities sum to 1; that is,

$$\sum i=0\infty p(i)=\sum i=0n(ni)pi(1-p)n-i=[p+(1-p)]n=1$$

Example 6a

Five fair coins are flipped. If the outcomes are assumed independent, find the probability mass function of the number of heads obtained.

Solution

If we let X equal the number of heads (successes) that appear, then X is a binomial random variable with parameters (n=5, p=12). Hence, by **Equation (6.2)**,

$$P\{ X=0 \}=(50)(12)0(12)5=132P\{ X=1 \}=(51)(12)1(12)4=532P\{ X=2 \}=(52)(12)2(12)3=1032P\{ X=3 \}=(53)(12)3(12)2=1032P\{ X=4 \}=(54)(12)4(12)1=532P\{ X=5 \}=(55)(12)5(12)0=132$$

Example 6b

It is known that screws produced by a certain company will be defective with probability .01, independently of one another. The company sells the screws in packages of 10 and offers a money-back guarantee that at most 1 of the 10 screws is defective. What proportion of packages sold must the

company replace?

Solution

If X is the number of defective screws in a package, then X is a binomial random variable with parameters (10, .01). Hence, the probability that a package will have to be replaced is

$$1-P\{ X=0 \}-P\{ X=1 \}=1-(100)(.01)0(.99)10-(101)(.01)1(.99)9\approx.004$$

Thus, only .4 percent of the packages will have to be replaced.

Example 6c

The following gambling game, known as the wheel of fortune (or chuck-a-luck), is quite popular at many carnivals and gambling casinos: A player bets on one of the numbers 1 through 6. Three dice are then rolled, and if the number bet by the player appears i times, i=1,2,3, then the player wins i units; if the number bet by the player does not appear on any of the dice, then the player loses 1 unit. Is this game fair to the player? (Actually, the game is played by spinning a wheel that comes to rest on a slot labeled by three of the numbers 1 through 6, but this variant is mathematically equivalent to the dice version.)

Solution

If we assume that the dice are fair and act independently of one another, then the number of times that the number bet appears is a binomial random variable with parameters (3,16). Hence, letting X denote the player's winnings in the game, we have

$$P\{ X=-1 \}=(30)(16)0(56)3=125216P\{ X=1 \}=(31)(16)1(56)2=75216P\{ X=2 \}=(32)(16)2(56)1=15216P\{ X=3 \}=(33)(16)3(56)0=1216$$

In order to determine whether or not this is a fair game for the player, let us calculate E[X]. From the preceding probabilities, we obtain

$$E[X]=-125+75+30+3216=-17216$$

Hence, in the long run, the player will lose 17 units per every 216 games he plays.

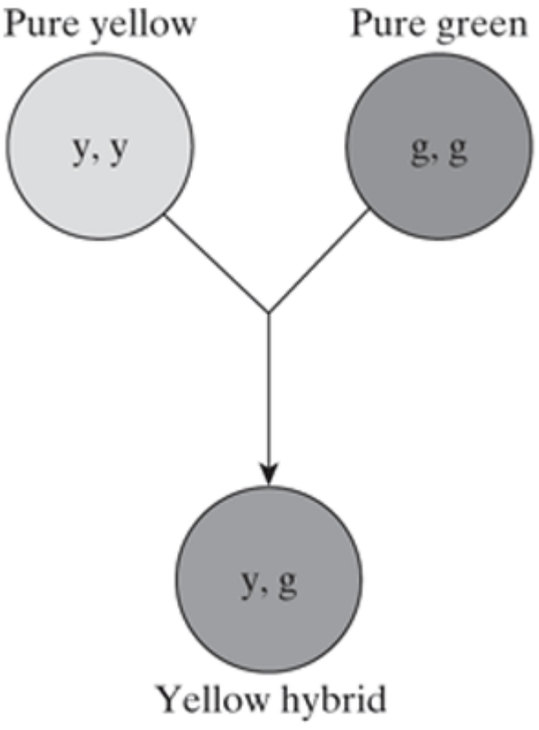
In the next example, we consider the simplest form of the theory of inheritance as developed by Gregor Mendel (1822—1884).

Example 6d

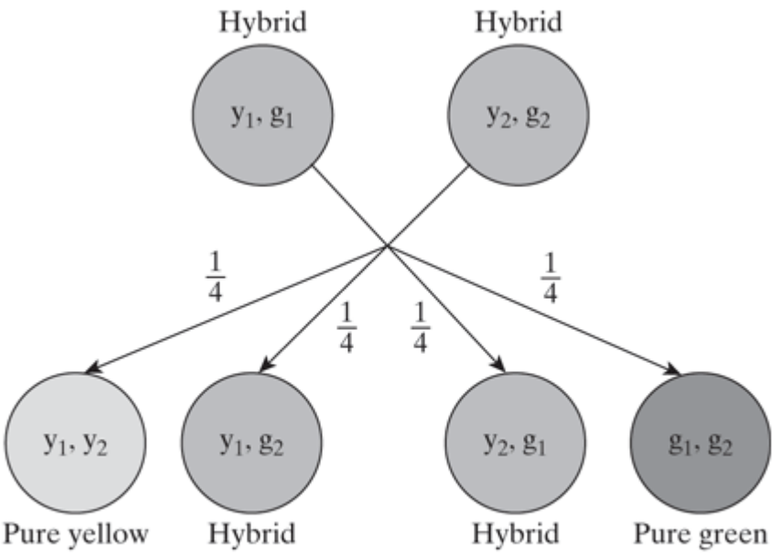
Suppose that a particular trait (such as eye color or left-handedness) of a person is classified on the basis of one pair of genes, and suppose also that d represents a dominant gene and r a recessive gene. Thus, a person with *dd* genes is purely dominant, one with *rr* is purely recessive, and one with *rd* is hybrid. The purely dominant and the hybrid individuals are alike in appearance. Children receive 1 gene from each parent. If, with respect to a particular trait, 2 hybrid parents have a total of 4 children, what is the probability that 3 of the 4 children have the outward appearance of the dominant gene?

The preceding **Figure 4.6a** and **b** shows what can happen when hybrid yellow (dominant) and green (recessive) seeds are crossed.

Figure 4.6 (a) Crossing pure yellow seeds with pure green seeds; (b) Crossing hybrid first-generation seeds.



(a)



(b)

Solution

If we assume that each child is equally likely to inherit either of 2 genes from each parent, the probabilities that the child of 2 hybrid parents will have *dd, rr*, and *rd* pairs of genes are, respectively, 14,14, and 12. Hence, since an offspring will have the outward appearance of the dominant gene if its gene pair is either *dd* or *rd*, it follows that the number of such children is binomially distributed with parameters (4,34). Thus, the desired probability is

(43)(34)3(14)1=2764

Example 6e

Consider a jury trial in which it takes 8 of the 12 jurors to convict the defendant; that is, in order for the defendant to be convicted, at least 8 of the jurors must vote him guilty. If we assume that jurors act independently and that whether or not the defendant is guilty, each makes the right decision with probability θ, what is the probability that the jury renders a correct decision?

Solution

The problem, as stated, is incapable of solution, for there is not yet enough information. For instance, if the defendant is innocent, the probability of the jury rendering a correct decision is

$$\sum_{i=5}^{12}\binom{12}{i}\theta^i(1-\theta)^{12-i}$$

whereas, if he is guilty, the probability of a correct decision is

$$\sum_{i=8}^{12}\binom{12}{i}\theta^i(1-\theta)^{12-i}$$

Therefore, if α represents the probability that the defendant is guilty, then, by conditioning on whether or not he is guilty, we obtain the probability that the jury renders a correct decision:

$$\alpha\sum_{i=8}^{12}\binom{12}{i}\theta^i(1-\theta)^{12-i}+(1-\alpha)\sum_{i=5}^{12}\binom{12}{i}\theta^i(1-\theta)^{12-i}$$

Example 6f

A communication system consists of n components, each of which will, independently, function with probability p. The total system will be able to operate effectively if at least one-half of its components function.

a. For what values of p is a 5-component system more likely to operate effectively than a 3-component system?
b. In general, when is a (2k+1)-component system better than a (2k−1)-component system?

Solution

a. Because the number of functioning components is a binomial random variable with parameters (n, p), it follows that the probability that a 5-component system will be effective is
$$\binom{5}{3}p^3(1-p)^2+\binom{5}{4}p^4(1-p)+p^5$$

   whereas the corresponding probability for a 3-component system is
$$\binom{3}{2}p^2(1-p)+p^3$$

   Hence, the 5-component system is better if
$$10p^3(1-p)^2+5p^4(1-p)+p^5>3p^2(1-p)+p^3$$

   which reduces to
$$3(p-1)^2(2p-1)>0$$

   or
$$p>\frac{1}{2}$$

b. In general, a system with 2k+1 components will be better than one with 2k−1 components if (and only if) p>12. To prove this, consider a system of 2k+1 components and let X denote the number

of the first 2k−1 that function. Then
P2k+1(effective)=P{ X≥k+1 }+P{ X=k }(1−(1−p)2)+P{ X=k−1 }p2

which follows because the (2k+1)-component system will be effective if either

i. X≥k+1;
ii. X=k and at least one of the remaining 2 components function; or
iii. X=k−1 and both of the next 2 components function.

Since
P2k−1(effective)=P{ X≥k }=P{ X=k }+P{ X≥k+1 }

we obtain
P2k+1(effective)−P2k−1(effective) =P{ X=k−1 }p2−(1−p)2P{ X=k
} =(2k−1k−1)pk−1(1−p)kp2−(1−p)2(2k−1k)pk(1−p)k−1 =(2k−1k)pk(1−p)k[p−(1−p)
] since(2k−1k−1)=(2k−1k) >0⇔p>12

4.6.1 Properties of Binomial Random Variables

We will now examine the properties of a binomial random variable with parameters n and p. To begin, let us compute its expected value and variance. To begin, note that

E[Xk]=∑i=0nik(ni)pi(1−p)n−i=∑i=1nik(ni)pi(1−p)n−i

Using the identity

i(ni)=n(n−1i−1)

gives

E[Xk]=np∑i=1nik−1(n−1i−1)pi−1(1−p)n−i=np∑j=0n−1(j+1)k−1(n−1j)pj(1−p)n−1−jby lettingj=i−1 =npE[
(Y+1)k−1]

where Y is a binomial random variable with parameters n−1, p. Setting k=1 in the preceding equation yields

E[X]=np

That is, the expected number of successes that occur in n independent trials when each is a success with probability p is equal to *np*. Setting k=2 in the preceding equation and using the preceding formula for the expected value of a binomial random variable yields

E[X2]=npE[Y+1]=np[(n−1)p+1]

Since E[X]=np, we obtain

Var(X)=E[X2]−(E[X])2=np[(n−1)p+1]−(np)2=np(1−p)

Summing up, we have shown the following:

If X is a binomial random variable with parameters n and p, then

E[X]=npVar(X)=np(1−p)

The following proposition details how the binomial probability mass function first increases and then decreases.

Proposition 6.1

If X is a binomial random variable with parameters (n, p), where 0<p<1, then as k goes from 0 to n,P{X=k} first increases monotonically and then decreases monotonically, reaching its largest value when k is the largest integer less than or equal to (n+1)p.

Proof We prove the proposition by considering P{X=k}/P{X=k−1} and determining for what values of k it is greater or less than 1. Now,

$$P\{X=k\}P\{X=k-1\}=n!(n-k)!k!pk(1-p)n-kn!(n-k+1)!(k-1)!pk-1(1-p)n-k+1=(n-k+1)pk(1-p)$$

Hence, $P\{X=k\}\geq P\{X=k-1\}$ if and only if

$$(n-k+1)p\geq k(1-p)$$

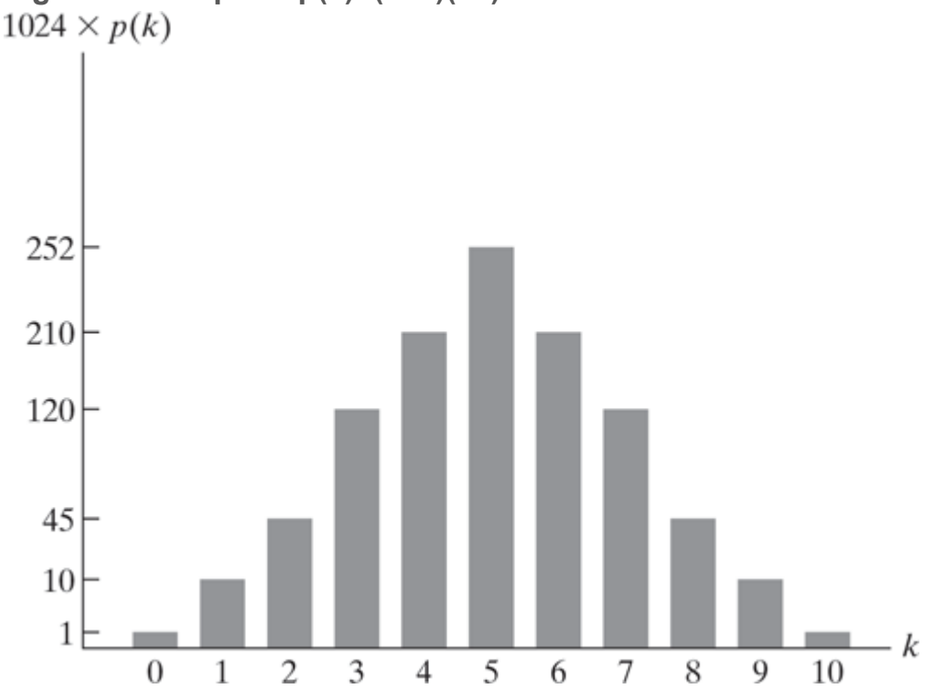
or, equivalently, if and only if

$$k\leq(n+1)p$$

and the proposition is proved.

As an illustration of **Proposition 6.1**, consider **Figure 4.7**, the graph of the probability mass function of a binomial random variable with parameters (10,12).

Figure 4.7 Graph of p(k)=(10k)(12)10.



Example 6g

In a U.S. presidential election, the candidate who gains the maximum number of votes in a state is awarded the total number of electoral college votes allocated to that state. The number of electoral college votes of a given state is roughly proportional to the population of that state—that is, a state with population n has roughly *nc* electoral votes. (Actually, it is closer to nc+2, as a state is given an electoral

vote for each member it has in the House of Representatives, with the number of such representatives being roughly proportional to the population of the state, and one electoral college vote for each of its two senators.) Let us determine the average power of a citizen in a state of size n in a close presidential election, where, by *average power in a close election*, we mean that a voter in a state of size n=2k+1 will be decisive if the other n−1 voters split their votes evenly between the two candidates. (We are assuming here that n is odd, but the case where n is even is quite similar.)

Because the election is close, we shall suppose that each of the other n−1=2k voters acts independently and is equally likely to vote for either candidate. Hence, the probability that a voter in a state of size n=2k+1 will make a difference to the outcome is the same as the probability that 2k tosses of a fair coin land heads and tails an equal number of times. That is,

$$P\{\text{ voter in state of size } 2k+1 \text{ makes a difference }\} = (2kk)(12)k(12)k = (2k)!k!k!22k$$

To approximate the preceding equality, we make use of Stirling's approximation, which says that for k large,

$$k! \sim kk+1/2e-k2\pi$$

where we say that $ak \sim bk$ when the ratio ak/bk approaches 1 as k approaches ∞. Hence, it follows that

$$P\{\text{ voter in state of size } 2k+1 \text{ makes a difference }\} \sim (2k)2k+1/2e-2k2\pi k2k+1e-2k(2\pi)22k = 1k\pi$$

Because such a voter (if he or she makes a difference) will affect *nc* electoral votes, the expected number of electoral votes a voter in a state of size n will affect—or the voter's average power—is given by

$$\text{average power} = ncP\{\text{ makes a difference }\} \sim ncn\pi/2 = c2n/\pi$$

Thus, the average power of a voter in a state of size n is proportional to the square root of n, showing that in presidential elections, voters in large states have more power than do those in smaller states.

4.6.2 Computing the binomial distribution function

Suppose that X is binomial with parameters (n, p). The key to computing its distribution function

$$P\{X \le i\} = \sum_{k=0}^{i} \binom{n}{k} p^k (1-p)^{n-k} \quad i = 0, 1, \ldots, n$$

is to utilize the following relationship between P{X=k+1} and P{X=k}, which was established in the proof of **Proposition 6.1**:

$$P\{X=k+1\} = p1-pn-kk+1P\{X=k\} \tag{6.3}$$

Example 6h

Let X be a binomial random variable with parameters n=6, p=.4. Then, starting with P{X=0}=(.6)6 and recursively employing **Equation (6.3)**, we obtain

$$P\{X=0\} = (.6)6 \approx .0467 \quad P\{X=1\} = 4661P\{X=0\} \approx .1866 \quad P\{X=2\} = 4652P\{X=1\} \approx .3110 \quad P\{X=3\} = 4643P\{X=2\} \approx .2765 \quad P\{X=4\} = 4634P\{X=3\} \approx .1382 \quad P\{X=5\} = 4625P\{X=4\} \approx .0369 \quad P\{X=6\} = 4616P\{X=5\} \approx .0041$$

A computer program that utilizes the recursion (**6.3**) to compute the binomial distribution function is easily written. To compute P{X≤i}, the program should first compute P{X=i} and then use the recursion to successively compute P{X=i−1},P{X=i−2}, and so on.

Historical note

Independent trials having a common probability of success p were first studied by the Swiss mathematician Jacques Bernoulli (1654–1705). In his book *Ars Conjectandi (The Art of Conjecturing)*, published by his nephew Nicholas eight years after his death in 1713, Bernoulli showed that if the number of such trials were large, then the proportion of them that were successes would be close to p with a probability near 1.

Jacques Bernoulli was from the first generation of the most famous mathematical family of all time. Altogether, there were between 8 and 12 Bernoullis, spread over three generations, who made fundamental contributions to probability, statistics, and mathematics. One difficulty in knowing their exact number is the fact that several had the same name. (For example, two of the sons of Jacques's brother Jean were named Jacques and Jean.) Another difficulty is that several of the Bernoullis were known by different names in different places. Our Jacques (sometimes written Jaques) was, for instance, also known as Jakob (sometimes written Jacob) and as James Bernoulli. But whatever their number, their influence and output were prodigious. Like the Bachs of music, the Bernoullis of mathematics were a family for the ages!

Example 6i

If X is a binomial random variable with parameters n=100 and p=.75, find P{X=70} and P{X≤70}.

Solution

A binomial calculator can be used to obtain the following solutions:

Figure 4.8

